## * Revision:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A) * P(B|A)$$

$$= P(B) * P(A|B)$$

→ **Random Experiment:**

A process whose result can not be predicted with certainty, even though all possible results are known in advance.

→ **Independent Events:**

$$P(A|B) = P(A)$$

$$P(A \cap B) = P(A) * P(B)$$

→ **Likelihood:**

$$L(\theta | data) = P(data | \theta)$$

- Probabity of data given some parameter value. (evidance)

Ex: $P(H) = 0.7$ — If we do three Independent trials then

↓

P(HHT | P(H)=0.7)

↳ L(P(H)=0.7 | HHT)

→ Bayes' Theorem: → P(θ | data)

P(data | θ)

↓

Posterior

P(H) → Experiment → Evidance → P(H|E) ?

$$P(H|E) = \frac{P(H \cap E)}{P(E)} = \frac{P(E|H) * P(H)}{P(E)}$$

Posterior ← Prior

Likelihood

P(E) ↳ Evidance (marginal likelihood)

## 8.4 The four ingredients

| Name | Symbol | Meaning in words |
|---|---|---|
| Prior | P(H) | What we believed about the hypothesis H before seeing any evidence. |
| Likelihood | P(E \| H) | How probable the observed evidence E would be if H were true. |
| Evidence (marginal likelihood) | P(E) | How probable the evidence is overall, across all hypotheses. |
| Posterior | P(H \| E) | Our updated belief about H after seeing the evidence. This is what we actually want. |

Ex In a school;

20% → play cricket

80% Stu → who play cricket → boys

40% " → " do not play cricket → boys

is a boy what is

If Randomaly selected stu. [illegible], the pro. that he plays cricket ?

→ $P(C) = 0.2$

$P(B/C) = 0.8$

$P(B/C) = 0.2$

$P(B/NC) = 0.4$

$P(NC) = 0.8$

$P(C/B)$ (?)

$$P(C/B) = \frac{P(B/C) * P(C)}{P(B)}$$

$$P(B) = P(B \cap C) + P(B \cap NC)$$

$$= P(B/C) * P(C) + P(B/NC) * P(NC)$$

$$= 0.16 + 0.32$$

$$= 0.48$$

$$P(C/B) = \frac{P(B/C) * P(C)}{P(B)}$$

$$= \frac{0.8 * 0.2}{0.48} = 0.333$$

Probability increases to 33% from 20% after considering the evidance.

* Today's class:

# Statistics

→ What is the Relationship of stastics with ml ?

* Population vs sample:

If we want to do a survey on students of India Lets say : Avg IQ of Indian students ?

↓

Is it possible ?

→ Practical Example: Exit poll

Is it always correct ?

→ Pop: Refers to the entire set of individuals, objects or data points that we want to study.

→ sample: Subset of population.

* Types of statistics:

Descriptive

↓

Summarizing & describing the main features of data without making any prediction or Inferences.

Eg: Avg

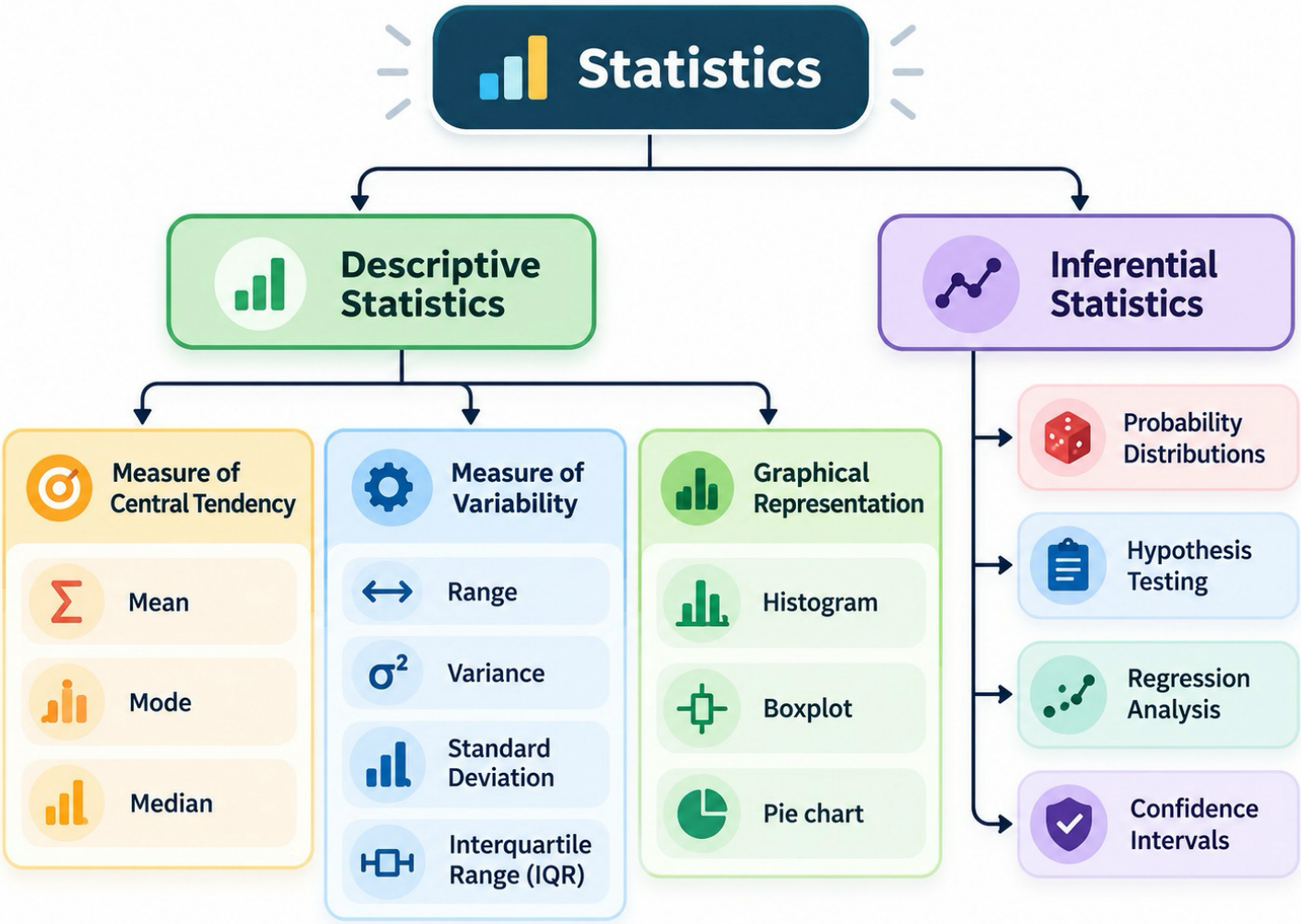
Inferential

↓

- Making conclusion
- Predictions

Eg: Exit poll

Prediction about a population based on a sample.


Statistics
Descriptive Statistics
Inferential Statistics
Measure of Central Tendency
Mean
Mode
Median
Measure of Variability
Range
Variance
Standard Deviation
Interquartile Range (IQR)
Graphical Representation
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals

Stastics it self is a big subject.

## * Measure of central Tendency:

- Represent the middle or single value that in a dataset.

① Mean (AVG): ② Median ③ Mode

② Median → middle value (data should be sorted)

③ Mode → most frequent value.

Ex: 1 1 1 4 5 6 7 8

Avg = $\frac{\text{Sum}}{8}$

middle value = $\frac{4+5}{2} = 4.5$

or

You can consider 4 or 5.

mode = 1

② 9 9 8 7 8 10 12 15 12 13

mean =

median =

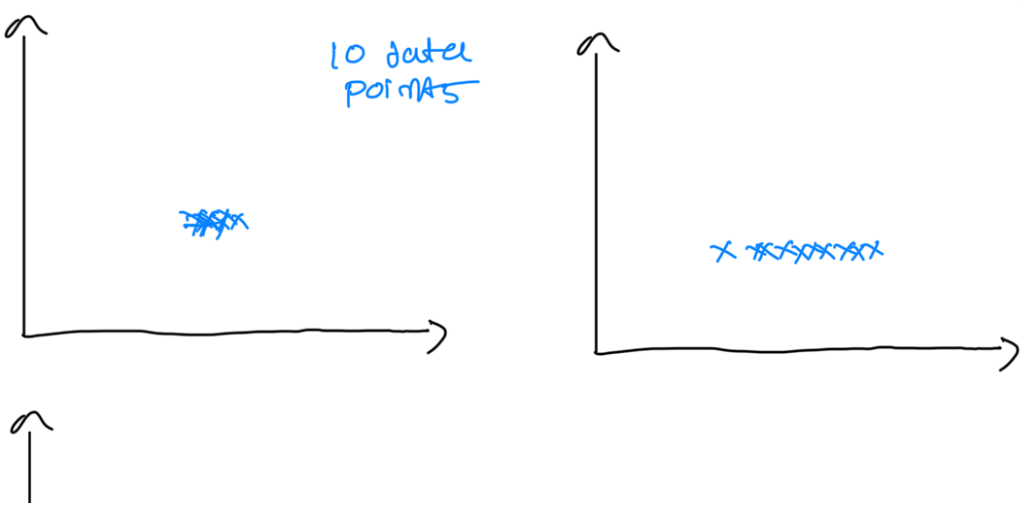
mode =

→ Now what if one student got placement of 80 LPA ?

Mean ? Median ? mode ?

→ Effect of outliers

* Measure of Dispersion: Spread
(फैलाव)



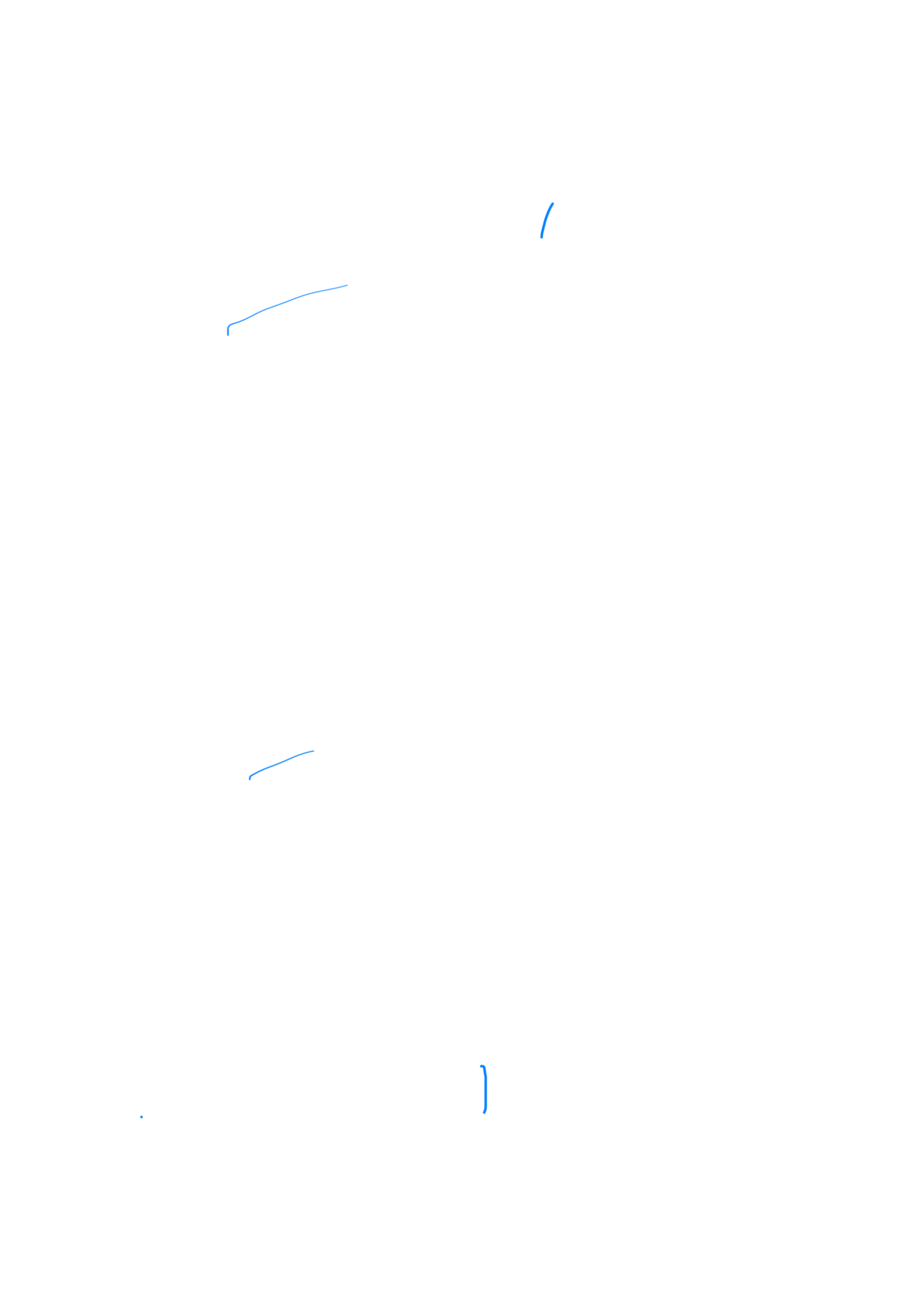


① Range:

Ex: 1 1 1 2 3 4 5 5 6 8 9 10

Range: 1 to 10

② Variance: It tells how far on avg. each no. in our dataset from mean.



⇒ defination 2: Proportion (percentage) of the total variance in the target explained by the model.

($R^2$ score)

$R^2 = 0.8$ : 80 % variance can be explained by our model.